

Diagnostic Platform and Oracle Database – Test Result Data Flow

Table of Contents

Purpose	1
Scope	1
System Overview	1
Data Flow Summary	2
Detailed Workflow	3
Data Flow Diagram.....	4
Compliance & Audit Traceability	5
Change Management.....	5
Appendix	5

1. Purpose

This document describes the data flow and process logic for transmitting diagnostic test results from the Siemens diagnostic platform to the Oracle backend database for secure storage and retrieval. It ensures data integrity, traceability, and compliance with FDA 21 CFR Part 11 and ISO 13485 requirements.

2. Scope

This document covers:

- Data extraction, transformation, and loading (ETL) from the diagnostic analyzer.
- Validation, encryption, and transfer protocols.
- Archival and audit logging workflows.
- Applicable for the Atellica and Dimension Vista systems integrated with the corporate Oracle DB.

3. System Overview

- Upstream Systems: Diagnostic Instrument Software (Atellica), Middleware Layer (Result Processing Server)
- Downstream Systems: Oracle Database (Production schema), Data Archive Server, Validation and Audit Log Repository

Author: Manoj Thakur	Version: 1.1	Date: Mar 2022
Reviewed by: QA Lead – Validation Systems	Confidential – Siemens Healthineers Internal Use Only	Page 1 of 6

4. Data Flow Summary

Step	Source	Destination	Description	Responsible Role	Validation Point
1	Diagnostic Analyzer	Middleware Server	Analyzer generates patient result in XML format	Lab System Engineer	Analyzer log
2	Middleware Server	Validation Module	XML validated against schema and QC thresholds	QA Validation System	Error log
3	Validation Module	Oracle Database	Validated results transformed to CSV and loaded via SQL Loader	Data Integration Engineer	ETL success log
4	Oracle Database	Archive Server	Records older than 90 days moved to cold storage	DBA	Archive log
5	Oracle DB	Audit Log Repository	All insert/update/delete events logged with user/time stamp	Compliance Team	Audit report

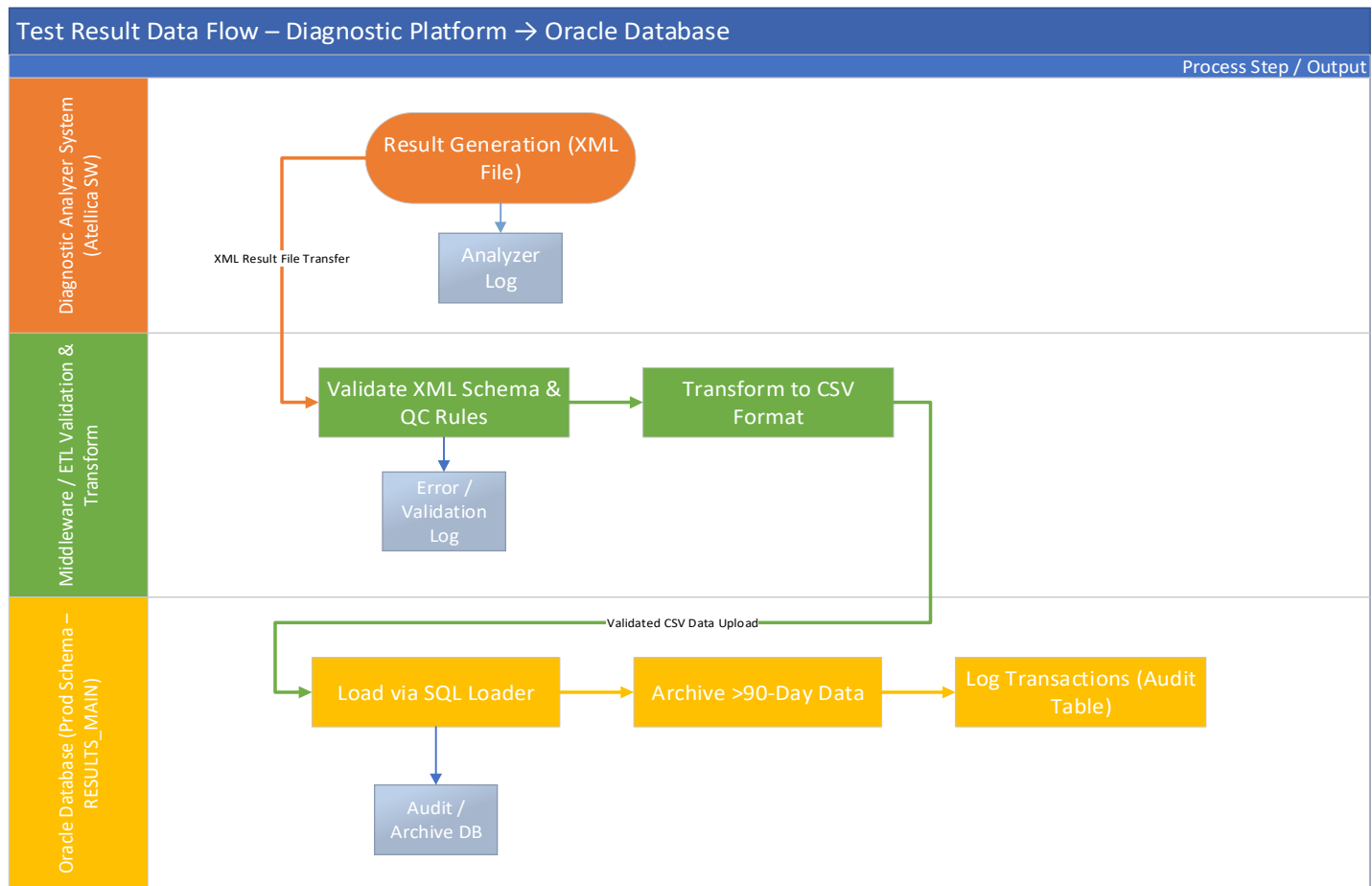
Author: Manoj Thakur	Version: 1.1	Date: Mar 2022
Reviewed by: QA Lead – Validation Systems	Confidential – Siemens Healthineers Internal Use Only	Page 2 of 6

5. Detailed Workflow

1. Result Generation: The diagnostic analyzer processes patient samples and generates an XML file containing test results.
2. File Validation and Transformation: Middleware validates the XML structure and converts valid results into CSV format.
3. Data Loading (ETL): SQL Loader inserts data into RESULTS_MAIN table with transaction logs.
4. Archival and Audit Trail: Records older than 90 days move to RESULTS_ARCHIVE with full audit logging.
5. Error Handling: Failed ETL transactions are retried twice and logged into ERROR_QUEUE for manual review.

Author: Manoj Thakur	Version: 1.1	Date: Mar 2022
Reviewed by: QA Lead – Validation Systems	Confidential – Siemens Healthineers Internal Use Only	Page 3 of 6

6. Data Flow Diagram



Author: Manoj Thakur	Version: 1.1	Date: Mar 2022
Reviewed by: QA Lead – Validation Systems	Confidential – Siemens Healthineers Internal Use Only	Page 4 of 6

7. Compliance & Audit Traceability

Compliance Area	Control Mechanism	Evidence
FDA 21 CFR Part 11	Electronic records validated via unique user authentication	User logs in AUDIT_LOGS
ISO 13485 Clause 7.5	Data integrity checks and validation schema	XML schema files and test scripts
Data Retention	Archive policy (90 days)	Archive job log
Change Control	Revision control in Veeva Vault	Doc revision history

8. Change Management

Version	Date	Author	Description	Reviewer
1.0	Jan 2022	Manoj Thakur	Initial draft for data flow mapping	QA Lead
1.1	Mar 2022	Manoj Thakur	Added archive logic and audit schema	Compliance Officer

Author: Manoj Thakur	Version: 1.1	Date: Mar 2022
Reviewed by: QA Lead – Validation Systems	Confidential – Siemens Healthineers Internal Use Only	Page 5 of 6

9. Appendix

- File schema samples (result.xsd, result_sample.xml)
- Oracle ETL script snippet
- Middleware validation configuration

Author: Manoj Thakur	Version: 1.1	Date: Mar 2022
Reviewed by: QA Lead – Validation Systems	Confidential – Siemens Healthineers Internal Use Only	Page 6 of 6